

# Toxic Comment Classification Using Machine Learning

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## Abstract

This project presents the development and evaluation of two machine learning models for classifying toxic comments. A logistic regression model using TF-IDF features and a fine-tuned BERT model were implemented. The models were trained on a dataset of 4000 comments from social media platforms. The fine-tuned BERT model outperformed logistic regression in terms of classification performance.

## 1. Introduction and Background

Toxic comments in social media create a hostile environment, discouraging constructive discussions. Automatic classification of such comments can help mitigate online toxicity. In this assignment, two models were developed: a fine-tuned BERT model and a traditional ML model (such as Logistic Regression or Random Forest) to classify comments as toxic or non-toxic.

This project builds on foundational work in toxicity detection, particularly the Jigsaw Toxic Comment Classification Challenge (2018), which established benchmark models using Wikipedia comments. However, social media toxicity presents unique challenges due to platform-specific slang (e.g., Twitter "ratioing"), abbreviated syntax, and multimedia context dependencies.

## 2. Dataset and Methods

### 2.1 Dataset

A dataset of 4000 comments sourced from Reddit, Twitter/X, and YouTube was used, which were human-annotated for toxicity.

#### Training Data

- Source: 4,000 comments from Reddit, Twitter, and YouTube.
- Labels: Composite toxicity derived from five human annotations via majority voting ( $\geq 3/5$  votes for toxicity).
- Annotation Consensus: 72% unanimous votes, 28% split decisions
- Features: Text body, parent comment, article title, platform metadata, and worker IDs.

## Test Data

- Size: 1000 Comments as unlabeled dataset with identical structure to training data.
- Platform Distribution: Mirrors training set ( $\chi^2$   $p=0.42$ )
- Key Characteristics:
  - Platform distribution: 45% Reddit, 35% YouTube, 20% Twitter.
  - Average text length: 28 words (range: 2–189).

## **2.2 Data processing**

The data preprocessing steps included:

1. The dataset was loaded from JSON format.
2. Text cleaning involved converting to lowercase, removing special characters, and normalizing whitespace.
3. TF-IDF vectorization was applied to extract features for the logistic regression model.
4. Tokenization and input encoding were performed for the BERT model using the Hugging Face tokenizer.

## **2.3 ML methods**

### **Logistic Regression Model**

- TF-IDF vectorization was used for text representation.
- The dataset was split into training and validation sets.
- The model was trained using Scikit-learn's `LogisticRegression`.
- Accuracy, precision, recall, and F1-score were used for evaluation.

### **Fine-Tuned BERT Model**

- A pre-trained BERT model (`BertForSequenceClassification`) was fine-tuned.
- Data was tokenized and converted into a PyTorch dataset.
- Hugging Face's `Trainer` API was used for model training.
- The model was evaluated using accuracy, precision, recall, and F1-score.

### 3. Evaluations and Findings

#### 3.1 Metrics

| Model               | Accuracy | Precision | Recall  | F1-Score |
|---------------------|----------|-----------|---------|----------|
| Logistic Regression | 0.77125  | 0.72917   | 0.17073 | 0.27668  |
| Fine-Tuned BERT     | 0.815    | 0.64322   | 0.62439 | 0.63366  |

#### 3.2 Key Findings

- BERT outperforms Logistic Regression in accuracy, recall, and F1-score, indicating superior overall classification capability and better handling of imbalanced data.
- Logistic Regression struggles with recall (0.171), suggesting difficulty identifying true positive cases in the dataset.
- BERT trades precision for recall: While BERT’s precision is lower (0.643 vs. 0.729), its recall is 3.6x higher, making it better suited for applications where false negatives are costly.

#### 3.3 Limitations

##### Logistic Regression

- Linearity assumption: Fails to model complex, non-linear relationships in text data, leading to severe underperformance in recall.
- Class imbalance sensitivity: Poor recall (17%) indicates challenges with skewed datasets unless explicitly adjusted.
- Feature dependency: Relies on manual feature engineering (e.g., TF-IDF), which may miss contextual nuances.

##### 2. Fine-Tuned BERT

- Computational cost: Requires significant resources for training and inference compared to simpler models.
- Overfitting risk: Validation loss increased from 0.422 (Epoch 1) to 0.721 (Epoch 3), suggesting potential overfitting despite early stopping.
- Contextual ambiguity: While BERT improves recall, its precision drop reflects residual challenges with subtle linguistic distinctions.

## Conclusion:

The Fine-Tuned BERT model demonstrates superior performance over Logistic Regression for text classification tasks, achieving higher accuracy (0.815 vs. 0.771) and significantly improved recall (0.624 vs. 0.171) and F1-score (0.634 vs. 0.277), making it better suited for applications prioritizing the identification of true positives (e.g., detecting critical patterns in imbalanced datasets). While BERT's precision is slightly lower (0.643 vs. 0.729), its contextual understanding offsets this limitation, and batch inference optimizes computational efficiency during deployment. However, validation loss trends (rising from 0.422 to 0.721 over 3 epochs) highlight potential overfitting risks, suggesting the need for early stopping or additional regularization. For resource-constrained scenarios, Logistic Regression remains a viable baseline, but BERT's performance justifies its adoption in tasks requiring nuanced text analysis.